

PRAYAS JEE 3.0 2025

MATHEMATICS (Practice Sheet)

Trigonometric Functions

Q1 If A and B are complimentary angles, then

- (A) $(1 + \tan \frac{A}{2})(1 + \tan \frac{B}{2}) = 2$
 (B) $(1 + \cot \frac{A}{2})(1 + \cot \frac{B}{2}) = 2$
 (C) $(1 + \sec \frac{A}{2})(1 + \operatorname{cosec} \frac{B}{2}) = 2$
 (D) $(1 - \tan \frac{A}{2})(1 - \tan \frac{B}{2}) = 2$

Q2 If $A = \tan 6^\circ \tan 42^\circ$ and $B = \cot 66^\circ \cot 78^\circ$ then

- (A) $A = 2B$
 (B) $A = \frac{1}{3}B$
 (C) $A = B$
 (D) $3A = 2B$

Q3 The value of expression $\sum_{\theta=0}^8 \frac{1}{1+\tan^3(10\theta)^\circ}$ is equal to

- (A) 5
 (B) $\frac{21}{4}$
 (C) $\frac{14}{3}$
 (D) $\frac{9}{2}$

Q4 If $\tan \alpha, \tan \beta$ are the roots of the equation $x^2 + px + 1 = 0$, then the value of $\sin^2(\alpha + \beta) + p \sin(\alpha + \beta) + \cos(\alpha + \beta)$ is

- + $q \cos^2(\alpha + \beta)$
 (A) independent of p but dependent on q
 (B) independent of q but dependent on p
 (C) independent of both p and q
 (D) dependent on both p and q

Q5 If $\cos 5\theta = a \cos \theta + b \cos^3 \theta + c \cos^5 \theta + d$, then

- (A) $a = 20$
 (B) $b = -30$
 (C) $a + b + c = 2$
 (D) $a + b + c + d = 1$

Q6 In a circle of diameter 40 cm the length of a chord is 20 cm. The length of minor arc

corresponding to the chord, is equal to

- (A) $\frac{10\pi}{3}$ cm
 (B) $\frac{20\pi}{3}$ cm
 (C) $\frac{5\pi}{3}$
 (D) $\frac{30\pi}{3}$ cm

Q7 If the angles of a triangle are in ratio 3 : 4 : 5, then smallest angle in degrees and the greatest angle

in radians, are

- (A) $50^\circ, \frac{\pi}{12}$
 (B) $57^\circ, \frac{\pi}{12}$
 (C) $45^\circ, \frac{5\pi}{12}$
 (D) $85^\circ, \frac{\pi}{12}$

Q8 The real number of x and y satisfy the equation $xy = \sin(2t)$ and $\frac{x}{y} = \tan(t)$ where $0 < t < \frac{\pi}{2}$. The value of $x^2 + y^2$, is

- (A) $\sqrt{2}$ (B) 1
 (C) 2 (D) 4

Q9 Exact value of $\frac{\cos 21^\circ - \sin 21^\circ}{\sin^2 57^\circ - \sin^2 33^\circ}$ is equal to

- (A) 2
 (B) $-\sqrt{2}$
 (C) 1
 (D) $\sqrt{2}$

Q10 Exact value of $\sec 10^\circ - \tan 10^\circ - \tan 40^\circ$ is equal to

- (A) $\sqrt{3}$
 (B) 2
 (C) 0
 (D) $\frac{-1}{\sqrt{3}}$

Q11 Consider the statements about $y = \frac{\sin 3x}{\sin x}$, $\sin x \neq 0$ Which of the following is (are) correct?

- (A) The minimum value of y is (-1)



- (B) The maximum value of y is 3
 (C) The minimum value of y is NOT defined
 (D) The maximum value of y is NOT defined

Q12 If $\tan \theta + \cot(30^\circ - \theta) + \cot(150^\circ - \theta)$, then
 $= k \tan(a\theta)$

- (A) $k = 3$
 (B) $a = 3$
 (C) $k = 1$
 (D) $a = 1$

Q13 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and
 $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.
 Based on this information, answer the following
 questions.

$\sin A + 2 \sin 3A + \sin 5A$ is equal to

- (A) $4 \sin 3A \cos^2 A$
 (B) $4 \cos 3A \sin^2 A$
 (C) $4 \sin 3A \sin^2 A$
 (D) $\cos 3A \cos^2 A$

Q14 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and
 $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.
 Based on this information, answer the following
 questions. $\frac{\sin A + \sin 2A + \sin 4A + \sin 5A}{\cos A + \cos 2A + \cos 4A + \cos 5A}$ is equal to

- (A) $\cot 3A$
 (B) $\tan 3A$
 (C) $2 \cot 3A$
 (D) $2 \tan 3A$

Q15 $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ and
 $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.
 Based on this information, answer the following
 questions. $\frac{\cot 7A + \cos 3A - \cos 5A - \cos A}{\sin 7A - \sin 3A - \sin 5A + \sin A}$ is equal to

- (A) $\cot 2A$
 (B) $\tan 2A$
 (C) $2 \cot 2A$
 (D) $2 \tan 2A$

Q16 The minimum value of
 $\frac{\sec^4 \alpha}{\tan^2 \beta} + \frac{\sec^4 \beta}{\tan^2 \alpha}, \alpha, \beta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$ is

- (A) 4
 (B) 8
 (C) 1
 (D) 16

Q17 $\sqrt{3} \sin \frac{\pi}{9} (4 + \sec \frac{\pi}{9})$ is equal to

- (A) 3
 (B) $\frac{1}{3}$
 (C) -3
 (D) $-\frac{1}{3}$

Q18 If $\lambda \tan^2 x + \frac{\cot^2 x}{\lambda} = 2$ where
 $\lambda > 0$ & $x \in (0, \frac{\pi}{2})$ then

$\sin^2 x + 5 \cos^2 x + 3 \sin x \cos x$ is equal to

- (A) $\frac{1}{(\lambda+1)}$
 (B) $\frac{(1-4\lambda)}{(\lambda+1)}$
 (C) $\frac{1-4\lambda+3\sqrt{\lambda}}{(\lambda+1)}$
 (D) $\frac{1+5\lambda+3\sqrt{\lambda}}{(\lambda+1)}$

Q19 $\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + 8 \cot 24^\circ$
 $= \cot \theta^\circ (0$

$< \theta < 90^\circ)$, then

- (A) $\cot(10\theta)^\circ = 1$
 (B) $\cot(15\theta)^\circ = 1$
 (C) $\cot \theta^\circ = 0$
 (D) $\cot(15\theta)^\circ = \sqrt{3}$

Q20 If $\sin x + \sin^2 x = 1$, then the value of
 $\cos^2 x + \cos^4 x$
 is

- (A) 0
 (B) 2
 (C) 1
 (D) None of these

Q21 If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of
 $4 \cos \alpha + 5 \sin \alpha = 1$, then the value of $\tan \alpha$
 is

- (A) $\frac{10-\sqrt{10}}{12}$
 (B) $\frac{\sqrt{10}-10}{12}$
 (C) $\frac{\sqrt{10}-10}{6}$
 (D) $\frac{10-\sqrt{10}}{6}$

Q22 $2 \sin\left(\frac{\pi}{22}\right) \sin\left(\frac{3\pi}{22}\right) \sin\left(\frac{5\pi}{22}\right) \sin\left(\frac{7\pi}{22}\right) \sin\left(\frac{9\pi}{22}\right)$ is
 equal to

- (A) $\frac{3}{16}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) $\frac{9}{32}$

Q23 The value of $\cot\left(\frac{\pi}{24}\right)$ is:



- (A) $\sqrt{2} + \sqrt{3} + 2 - \sqrt{6}$
 (B) $\sqrt{2} + \sqrt{3} + 2 + \sqrt{6}$
 (C) $\sqrt{2} - \sqrt{3} - 2 + \sqrt{6}$
 (D) $3\sqrt{2} - \sqrt{3} - \sqrt{6}$

Q24 The value of

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}} \text{ is :}$$

- (A) $\frac{1}{256}$
 (B) $\frac{1}{2}$
 (C) $\frac{1}{512}$
 (D) $\frac{1}{1024}$

Q25 The value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$ is

- (A) $\frac{1}{36}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) $\frac{1}{18}$

Q26 Match the items of Column-I with those of Column II.

	Column I	Column II	
(A)	Absolute value of $288 \cos \frac{\pi}{31} \cos \frac{2\pi}{31} \cos \frac{4\pi}{31} \cos \frac{8\pi}{31} \cos \frac{16\pi}{31}$ is	(p)	4
(B)	The value of $\left[\tan 130^\circ - \tan 10^\circ + \sqrt{3} \tan 130^\circ \tan 10^\circ \right]^2$ is	(q)	2
(C)	If $\theta = \frac{2017\pi}{5}$, then the value of $2(\sin^2 4\theta + \cos^2 \theta)$ is	(r)	3
(D)	Value of expression $\frac{\sin 56^\circ + \sin 154^\circ + \sin 150^\circ}{\sin 28^\circ \sin 77^\circ \sin 75^\circ}$ is	(s)	9

- (A) (A) -s; (B) -r; (C) -q; (D) -p
 (B) (A) -p; (B) -q; (C) -r; (D) -s
 (C) (A) -s; (B) -r; (C) -p; (D) -q
 (D) (A) -r; (B) -s; (C) -q; (D) -p

Q27

Match the items of column-I with of column-II.

	Column I	Column II	
(A)	Value of $(\cos 24^\circ \cos 36^\circ \cos 84^\circ)$ is $\frac{\sqrt{a}-1}{b}$ (a and b are coprime numbers), then the value of $(a+b)$ is	(p)	12
(B)	Let $\alpha + \beta = \frac{\pi}{5}$, $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, if then the maximum value of $\sin \alpha + \sin \beta$ is $\frac{\sqrt{a}-1}{b}$, (a and b are coprime numbers), then the value of $(2a+b)$ is	(q)	21
(C)	Value of $\left(\cot \frac{\pi}{12} + \cot \frac{\pi}{8} - \tan \frac{\pi}{8} \right)$ is $(a + \sqrt{b})$ (a and b are coprime numbers), then the value of $(2a+3b)$ is	(r)	7
(D)	If the solution of $\cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 85^\circ = \frac{a}{b}$ (where a and b are prime numbers), then the value of $(a-5b)$ is	(s)	17

- (A) (A) -p; (B) -q; (C) -r; (D) -s
 (B) (A) -q; (B) -r; (C) -p; (D) -s
 (C) (A) -q; (B) -p; (C) -r; (D) -s
 (D) (A) -q; (B) -p; (C) -s; (D) -r

Q28 Match the items of column I with those of column II

	Column I	Column II	
(A)	$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to	(p)	0
(B)	If $\tan A + \tan B + \tan C = 6$ and $\tan A \tan B = 2$ where $A + B + C = 180^\circ$, then $\tan C$ is	(q)	$\sqrt{3}$
(C)	The value of $\cot 70^\circ + 4 \cos 70^\circ$ is	(r)	4
(D)	If $\tan^2 \theta = 2 \tan^2 \alpha + 1$, then the value of $\cos 2\theta + \sin^2 \alpha$ is	(s)	3

- (A) (A) -s; (B) -r; (C) -q; (D) -p
 (B) (A) -r; (B) -s; (C) -p; (D) -q
 (C) (A) -r; (B) -s; (C) -q; (D) -p
 (D) (A) -s; (B) -q; (C) -r; (D) -p



Q29 If the expression, $2 \cos 10^\circ + \sin 100^\circ + \sin 1000^\circ + \sin 10000^\circ$ is simplified, then it simplifies to a $\cos$ where a is natural number and b° is acute angle, then $b - a$ is

Q30 The number of integral values of n so that $\sin x(\sin x + \cos x) = n$ has at least one solution, is

Q31 The value of $36(4 \cos^2 9^\circ - 1)(4 \cos^2 27^\circ - 1)(4 \cos^2 81^\circ - 1)(4 \cos^2 243^\circ - 1)$ is
(A) 54 (B) 18
(C) 27 (D) 36

Q32 $96 \cos \frac{\pi}{33} \cos \frac{2\pi}{33} \cos \frac{4\pi}{33} \cos \frac{8\pi}{33} \cos \frac{16\pi}{33}$ is equal to
(A) 3 (B) 2
(C) 4 (D) 1

Q33 If $0 < x, y < \pi$ and $\cos x + \cos y - \cos(x + y) = 3/2$, Then, $\sin x + \cos y$ is equal to:
(A) $\frac{1+\sqrt{3}}{2}$
(B) $\frac{1-\sqrt{3}}{2}$
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{1}{2}$

Q34 The set of all values of λ for which the equation $\cos^2 2x - 2 \sin^4 x - 2 \cos^2 x = \lambda$
(A) $[-2, -1]$
(B) $[-2, -\frac{3}{2}]$
(C) $[-1, -\frac{1}{2}]$
(D) $[-\frac{3}{2}, -1]$

Q35 If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $(a + \frac{1}{a})$ is:
(A) 4
(B) $4 - 2\sqrt{3}$
(C) 2
(D) $5 - \frac{3}{2}\sqrt{3}$

Q36

If $L = \sin^2(\frac{\pi}{16}) - \sin^2(\frac{\pi}{8})$ and $M = \cos^2(\frac{\pi}{16}) - \sin^2(\frac{\pi}{8})$, then
(A) $L = -\frac{1}{2\sqrt{2}} + \frac{1}{2}\cos \frac{\pi}{8}$
(B) $M = \frac{1}{2\sqrt{2}} + \frac{1}{2}\cos \frac{\pi}{8}$
(C) $M = \frac{1}{4\sqrt{2}} + \frac{1}{4}\cos \frac{\pi}{8}$
(D) $L = \frac{1}{4\sqrt{2}} - \frac{1}{4}\cos \frac{\pi}{8}$

Q37 If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C = (x^{-3} + x^{-2} + x^{-1})^{1/2}$, $0 < A, B, C$, $< \frac{\pi}{2}$
then $A + B$ is equal to :
(A) C
(B) $\frac{\pi}{2} - C$
(C) $2\pi - C$
(D) $\pi - C$

Q38 The value of $2 \sin(\frac{\pi}{8}) \sin(\frac{2\pi}{8}) \sin(\frac{3\pi}{8}) \sin(\frac{5\pi}{8}) \sin(\frac{6\pi}{8}) \sin(\frac{7\pi}{8})$ is
(A) $\frac{1}{4\sqrt{2}}$
(B) $\frac{1}{4}$
(C) $\frac{1}{8}$
(D) $\frac{1}{8\sqrt{2}}$

Q39 If $15 \sin^4 \alpha + 10 \cos^4 \alpha = 6$, for some $\alpha \in R$, then the value of $27 \sec^6 \alpha + 8 \operatorname{cosec}^6 \alpha$ is equal to:
(A) 350 (B) 500
(C) 400 (D) 250

Q40 If $\cos(\alpha + \beta) = \frac{3}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to :
(A) $\frac{21}{16}$
(B) $\frac{63}{52}$
(C) $\frac{33}{52}$
(D) $\frac{63}{16}$

Q41 If α and β are acute angles satisfying $\cos 2\alpha = \frac{3 \cos 2\beta - 1}{3 - \cos 2\beta}$, then $\tan \alpha =$
(A) $\sqrt{2} \tan \beta$
(B) $\frac{1}{\sqrt{2}} \tan \beta$
(C) $\sqrt{2} \cot \beta$
(D)



$$\frac{1}{\sqrt{2}} \cot \beta$$

Q42 The radian measure corresponding to $-37^\circ 30'$ is equal to

- (A) $-\frac{5\pi}{24}$
 (B) $\frac{\pi}{24}$
 (C) $\frac{5\pi}{24}$
 (D) $-\frac{7\pi}{24}$

Q43 The expression,

$$\frac{\tan\left(\frac{3\pi}{2}-\alpha\right) \cos\left(\frac{3\pi}{2}-\alpha\right)}{\cos(2\pi-\alpha)} + \cos\left(\alpha - \frac{\pi}{2}\right) \sin(\pi - \alpha)$$

$$+ \cos(\pi + \alpha) \sin\left(\alpha - \frac{\pi}{2}\right), \alpha \neq \frac{n\pi}{2}, n \in I,$$

when simplified reduces to

- (A) 0 (B) 1
 (C) -1 (D) 2

Q44 The expression

$$f(x) = \frac{2 \sin^2 x - 1}{\cos x} + \frac{\cos x (1 + 2 \sin x)}{1 + \sin x}$$

defined simplifies to

- (A) $\tan^2 x$
 (B) $\tan x$
 (C) $1 - \cos x$
 (D) $1 + \sin^2 x$

Q45 If $A = 10^\circ$, then the value of expression

$$\left(\frac{\sec 2A}{(1 - 4 \sin^2 2A)} + \frac{\operatorname{cosec} A}{(4 \cos^2 A - 1)} \right)^2$$

- (A) 6 (B) 8
 (C) 16 (D) 24

Q46 The value of

$$\cos \frac{\pi}{65} \cos \frac{2\pi}{65} \cos \frac{4\pi}{65} \cos \frac{8\pi}{65} \cos \frac{16\pi}{65} \cos \frac{32\pi}{65}$$

- (A) $\frac{1}{8}$
 (B) $\frac{1}{16}$
 (C) $\frac{1}{32}$
 (D) none of these

Q47 The value of

$$\sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{\pi}{9}\right) + \sin^2\left(\frac{7\pi}{18}\right) + \sin^2\left(\frac{4\pi}{9}\right)$$

- (A) 1 (B) 2
 (C) 4 (D) none of these

Q48 The value of $\frac{\cot x - \tan x}{\cot 2x}$ is

Q49

The value of $\frac{2(\sin 2x + 2 \cos^2 x - 1)}{\cos x - \sin x - \cos 3x + \sin 3x}$ is

- (A) $\cos x$
 (B) $\sec x$
 (C) $\operatorname{cosec} x$
 (D) $\sin x$

Q50 The value of

$$2 \sin^2 B + 4 \cos(A + B) \sin A \sin B + \cos$$

$$2(A + B)$$

- (A) 0 (B) $\cos 3A$
 (C) $\cos 2A$ (D) none of these



Answer Key

Q1 (A)
Q2 (C)
Q3 (A)
Q4 (B)
Q5 (D)
Q6 (B)
Q7 (C)
Q8 (C)
Q9 (D)
Q10 (C)
Q11 (A, D)
Q12 (A, B)
Q13 (A)
Q14 (B)
Q15 (A)
Q16 (B)
Q17 (A)
Q18 (D)
Q19 (B)
Q20 (C)
Q21 (B)
Q22 (B)
Q23 (B)
Q24 (C)
Q25 (B)

Q26 (A)
Q27 (D)
Q28 (C)
Q29 9
Q30 2
Q31 (D)
Q32 (A)
Q33 (A)
Q34 (D)
Q35 (A)
Q36 (B)
Q37 (A)
Q38 (C)
Q39 (D)
Q40 (D)
Q41 (A)
Q42 (A)
Q43 (A)
Q44 (B)
Q45 (C)
Q46 (D)
Q47 (B)
Q48 2
Q49 (C)
Q50 (C)

